

Project 2: Wazuh SIEM Lab-EndPoint Security

Introduction

This project demonstrates the implementation of a Wazuh SIEM Home Lab for endpoint security monitoring and threat detection. The lab consists of an Ubuntu Server configured as the Wazuh Manager and a Windows machine configured as the Wazuh Agent. The Windows endpoint securely communicates with the Wazuh Manager, enabling centralized log collection, security event monitoring, and File Integrity Monitoring (FIM). The objective of this project is to understand how a SIEM solution collects, analyzes, and visualizes security events in real time through the Wazuh Dashboard while providing hands-on experience with endpoint monitoring and incident detection.

Lab Setup

- **Virtualization Platform:** VirtualBox
- **Wazuh Manager:** Ubuntu Server 22.04 LTS
- **Wazuh Agent:** Windows 10
- **Network Configuration:** Bridged Adapter (to enable communication between Ubuntu and Windows)
- **Components Installed:** Wazuh Manager, Wazuh Indexer, Wazuh Dashboard, and Wazuh Windows Agent
- **Purpose:** To build a SIEM home lab for centralized security monitoring, log analysis, and File Integrity Monitoring (FIM).

Prerequisites

- VirtualBox installed

- Ubuntu Server 22.04 LTS
- Windows 10
- Stable internet connection
- Basic Linux command-line knowledge

Tools Used

- Wazuh SIEM
- Ubuntu Server 22.04
- Windows 10
- VirtualBox
- Wazuh Dashboard
- Linux Terminal

Exercises

Exercise 1: Lab Setup

Objective: Set up the virtual environment for the Wazuh SIEM lab.

Step 1: Create the Ubuntu Virtual Machine

- Create an Ubuntu Server 22.04 LTS virtual machine in VirtualBox.

Step 2: Configure the Network

- Open Settings → Network.
- Set Adapter 1 to Bridged Adapter.

Step 3: Verify Internet Connectivity

[ping google.com](#)

Explanation: Verifies that the Ubuntu VM has internet access.

Step 4: Update the System

[sudo apt update && sudo apt upgrade -y](#)

Explanation: Updates the package list and installs the latest system updates before installing Wazuh.

Exercise 2: Install Wazuh Manager & Register Agent

Objective: Install the Wazuh Manager and register the Windows agent.

Step 1: Add the Wazuh GPG Key

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh-archive-keyring.gpg
```

Explanation: Adds the official Wazuh GPG key to verify package authenticity.

Step 2: Install Wazuh

```
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh && sudo bash ./wazuh-install.sh -a -i
```

Explanation: Installs the Wazuh Manager, Indexer, and Dashboard.

Step 3: Create a Windows Agent

```
sudo /var/ossec/bin/manage_agents
```

- Press **A** to add a new agent.
- Enter the **Agent Name**.
- Enter the **Windows IP Address** (or leave it blank).
- Save the agent.

Explanation: Creates a new Windows endpoint on the Wazuh Manager.

Step 4: Generate the Authentication Key

- Press **E** in the manage_agents menu.
- Enter the **Agent ID**.
- Copy the generated authentication key.

Explanation: Generates the key required to connect the Windows agent with the Wazuh Manager.

Exercise 3: Configure the Windows Agent

Objective: Connect the Windows agent to the Wazuh Manager.

Step 1: Install the Wazuh Agent

- Download and install the **Wazuh Windows Agent** using the default installation settings.

Step 2: Configure the Agent

- Open **Wazuh Agent Manager**.
- Enter the **Ubuntu Manager IP Address**.
- Paste the **Authentication Key** generated from the Ubuntu Manager.
- Click **Save**.

Explanation: Configures the Windows endpoint to communicate with the Wazuh Manager.

Step 3: Restart the Wazuh Agent

```
net stop WazuhSvc
```

```
net start WazuhSvc
```

Explanation: Restarts the Wazuh Agent service and applies the new configuration.

Exercise 4: Verify Agent Connection

Objective: Verify that the Windows agent is successfully connected to the Wazuh Manager.

Step 1: Check Agent Status

```
sudo /var/ossec/bin/agent_control -l
```

Explanation: Displays the list of registered agents and their current status.

Step 2: Verify in the Wazuh Dashboard

- Open the **Wazuh Dashboard** in your browser.
- Navigate to **Agents**.
- Confirm that the Windows agent is listed with the status **Active**.

Explanation: Confirms that the Windows endpoint is successfully communicating with the Wazuh Manager.

Exercise 5: Configure File Integrity Monitoring (FIM)

Objective: Configure Wazuh to monitor file changes on the Windows endpoint.

Step 1: Open the Configuration File

Navigate to:

[C:\Program Files \(x86\)\ossec-agent\ossec.conf](#)

Step 2: Add the Monitoring Directory

Add the following line inside the <syscheck> section:

[<directories realtime="yes">C:\Users\abc\Test</directories>](#)

Explanation: Enables real-time monitoring for the specified directory.

Step 3: Restart the Wazuh Agent

[net stop WazuhSvc](#)

[net start WazuhSvc](#)

Explanation: Restarts the Wazuh Agent to apply the File Integrity Monitoring configuration.

Exercise 6: Test File Integrity Monitoring

Objective: Verify that Wazuh detects file changes in real time.

Step 1: Perform File Operations

In the monitored folder:

- Create a new file.
- Modify the file.
- Delete the file.

Step 2: Verify Alerts

- Open the **Wazuh Dashboard**.
- Navigate to **Integrity Monitoring**.
- Confirm that alerts are generated for each file operation.

Explanation: Verifies that File Integrity Monitoring (FIM) is working correctly and that the Windows agent is sending events to the Wazuh Manager.

Exercise 7: Monitor Security Events in the Wazuh Dashboard

Objective: Analyze the security events generated by the Windows endpoint.

Step 1: Open the Dashboard

- Log in to the **Wazuh Dashboard**.
- Navigate to **Security Events** or **Threat Hunting**.

Step 2: Analyze Events

- View the events generated by the Windows agent.
- Filter events by **Agent**, **Rule Level**, or **Event Type**.
- Review the details of the detected activities.

Explanation: This exercise demonstrates how Wazuh collects, analyzes, and displays security events from the monitored endpoint through the centralized dashboard.

Exercise 8: Generate and Analyze Security Alerts

Objective: Generate security events and analyze the alerts in the Wazuh Dashboard.

Step 1: Trigger Security Events

- Perform activities such as creating, modifying, or deleting files in the monitored directory.
- Wait for the Windows agent to send the events to the Wazuh Manager.

Step 2: Review Alerts

- Open the **Wazuh Dashboard**.
- Navigate to **Security Events** or **Alerts**.
- Verify that the generated events are displayed with their corresponding rule level and description.

Explanation: This exercise validates that the Wazuh SIEM successfully detects, processes, and visualizes security events from the Windows endpoint.